

Distributions

Problem D.1

String with length L and a point mass at $x = a$

Solution:

$$\begin{aligned}\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} &= -\frac{m_0 g}{\rho} \delta(x - a), \quad 0 < x < L, t > 0 \\ u(0, t) &= u(L, t) = 0, \quad t > 0 \\ u(x, t) &= g(x), \quad \frac{\partial u(x, 0)}{\partial t} = h(x), \quad 0 < x < L\end{aligned}$$

□

Problem D.2

Show that

- (a) $g(x)\delta'(x) = g(0)\delta'(x) - g'(0)\delta(x)$
 (b) $g(x)\delta''(x) = g(0)\delta''(x) - 2g'(0)\delta'(x) + g''(0)\delta(x)$

Solution:

- (a) Here we use $g(x)\delta(x) = g(0)\delta(x)$, and start by differentiating the expression $(g(x)\delta(x))$:

$$\begin{aligned}(g(x)\delta(x))' &= g'(x)\delta(x) + g(x)\delta'(x) \\ &\downarrow \\ g(x)\delta'(x) &= (g(x)\delta(x))' - g'(x)\delta(x) = \\ &= (g(0)\delta(x))' - g'(0)\delta(x) = g(0)\delta'(x) - g'(0)\delta(x)\end{aligned}$$

(b) Using the result from (a) and differentiating $(g(x)\delta'(x))$:

$$\begin{aligned}
 (g(x)\delta'(x))' &= g'(x)\delta'(x) + g(x)\delta''(x) \\
 &\downarrow \\
 g(x)\delta''(x) &= (g(x)\delta'(x))' - g'(x)\delta'(x) = \\
 &= (g(0)\delta'(x) - g'(0)\delta(x))' - g'(x)\delta'(x) = \\
 &= g(0)\delta''(x) - g'(0)\delta'(x) - g'(0)\delta'(x) + g''(0)\delta(x) = \\
 &= g(0)\delta''(x) - 2g'(0)\delta'(x) + g''(0)\delta(x)
 \end{aligned}$$

□

Problem D.3

Simplify

(a) $e^x\delta_1$

(b) $x\delta'$

(c) $\cos(x)\delta'$

(d) $x^2\delta''$

(e) $\int g(t)\delta(t-1) dt$

Solution:

(a)

$$e^x\delta_1 = e^x\delta_1 = e\delta_1$$

(b) Using $g(x)\delta'(x) = g(0)\delta'(x) - g'(0)\delta(x)$

$$x\delta' = -\delta$$

(c) Using $g(x)\delta'(x) = g(0)\delta'(x) - g'(0)\delta(x)$:

$$\cos(x)\delta' = \cos(0)\delta' + \sin(0)\delta = \delta'$$

(d) Using $g(x)\delta''(x) = g(0)\delta''(x) - 2g'(0)\delta'(x) + g''(0)\delta(x)$:

$$x^2\delta'' = 2\delta$$

(c) Using $\int g(x)\delta(x-a) dx = g(a)$:

$$\int g(t)\delta(t-1) dt = g(1)$$

□

Problem D.4

Calculate f' and f'' in distributions for

(a) $f(x) = e^{-x}\theta(x)$

(b) $f(x) = e^{-|x|}$

Solution:

(a) Using $\theta'(x) = \delta(x)$

$$f'(x) = (e^{-x}\theta(x))' = -e^{-x}\theta(x) + e^{-x}\delta(x) = \delta(x) - e^{-x}\theta(x)$$

$$f''(x) = (\delta(x) - e^{-x}\theta(x))' = \delta'(x) + e^{-x}\theta(x) - e^{-x}\delta(x) = \delta'(x) - \delta(x) + e^{-x}\theta(x)$$

(b) Expressing $|x|$ in distributions:

$$|x| = \begin{cases} x, & x > 0 \\ -x, & x < 0 \end{cases} = x\theta(x) - x(1-\theta(x)) = 2x\theta(x) - x$$

$$f'(x) = (-2\theta(x) - 2x\delta(x) + 1)e^{-|x|} = (1 - 2\theta(x))e^{-|x|}$$

$$\begin{aligned} f''(x) &= (e^{-|x|} - 2e^{-|x|}\theta(x))' = (1 - 2\theta(x))e^{-|x|} + 2(1 - 2\theta(x))e^{-|x|}\theta(x) - 2e^{-|x|}\delta(x) = \\ &= e^{-|x|}[(1 - 2\theta(x)) + 2\theta(x)(1 - 2\theta(x))] - 2\delta(x) = e^{-|x|}(1 - 2\theta(x))^2 - 2\delta(x) = \\ &= [(1 - 2\theta(x))^2 = 1 \text{ for all } x \neq 0] = e^{-|x|} - 2\delta(x) \end{aligned}$$

□

Problem D.5

(a) Calculate the derivative of $x^2\theta(x-1)$

(b) Determine all primitive distributions for $x\theta(x-1)$

Solution:

(a)

$$(x^2\theta(x-1))' = 2x\theta(x-1) + x^2\theta'(x-1) = 2x\theta(x-1) + \delta(x-1)$$

(b) Try to compensate for the $\delta(x-1)$ in (a) by $x^2 \rightarrow x^2 - 1$:

$$((x^2 - 1)\theta(x-1))' = 2x\theta(x-1) + (x^2 - 1)\delta(x-1) = 2x\theta(x-1)$$

Now compensate with a factor 2 and we get the solution

$$\left(\frac{x^2 - 1}{2}\right)\theta(x-1) + C$$

□

Problem D.6

A string with length 1. A force F acts on the string at a point, distance a from the strings left end. The force is perpendicular to the string. No other external forces act on the string. Write out the differential equation and boundary conditions for the strings stationary and solve the problem.

Solution:

$$\frac{\partial^2 u}{\partial t^2} - S \frac{\partial^2 u}{\partial x^2} = F\delta(x-a)$$

$$u(0, t) = u(1, t) = 0$$

Since we are looking for the stationary solution, $\frac{\partial^2 u}{\partial t^2} = 0$ and we have

$$-\frac{\partial^2 u}{\partial x^2} = \frac{F}{S}\delta(x-a)$$

Integrating both sides twice gives

$$-u(x, t) = \frac{F}{S}(x-a)\theta(x-a) + Ax + B$$

Now check the boundary conditions for this expression:

$$u(0, t) = \frac{F}{S} \underbrace{(-a)\theta(-a)}_{=0 \text{ since } a>0} + B = 0 \rightarrow B = 0$$

$$u(1, t) = \frac{F}{S}(a-1)\theta(1-a) + A = 0 \rightarrow A = \frac{F}{S}(1-a)$$

$$u(x, t) = \frac{F}{S}((x(1-a) + (a-x)\theta(x-a)))$$

□

Problem D.7

Solve the equation

$$u'(x) + (2x+1)u(x) = \delta'(x)$$

Solution:

Integrating factor for LHS is

$$e^{x^2+x}$$

Multiply this into the equation and integrate LHS, then using the expansion for $f(x)\delta'(x)$:

$$\begin{aligned} (u(x)e^{x^2+x})' &= e^{x^2+x}\delta'(x) = e^0\delta'(x) - (2x+1)_{x=0}\delta(x) = \\ &= (u(x)e^{x^2+x})' = \delta'(x) - \delta(x) \end{aligned}$$

Now integrating both sides gives:

$$u(x)e^{x^2+x} = \delta(x) - \theta(x) + C$$

$$\downarrow$$

$$u(x) = e^{-(x^2+x)}(\delta(x) - \theta(x) + C) = \delta(x) + e^{-x^2-x}(C - \theta(x))$$

□

Problem D.8

Determine all distribution solution to

(a) $xU = 0$

- (b) $(x - 1)U = 0$
 (c) $(e^x - 2)U = 0$
 (d) $(x^2 - 1)U = 0$
 (e) $\sin(x)U = 0$
 (f) $x^3U = 0$

Solution:

(a)

$$U = \delta(x) \rightarrow x\delta(x) = 0 \rightarrow U = c\delta(x), \quad c = \text{constant}$$

(b) Same as in (a) but $\delta(x - 1)$

$$U = c\delta(x - 1)$$

(c) We can try to set U as $\delta(x - a)$ to make $(e^x - 2) = 0 \rightarrow e^x = 2 \rightarrow x = \ln(2)$

$$U = c\delta(x - \ln(2))$$

(d) Same idea as above, but here we have two roots, $x = 1, x = -1$

$$U = c_1\delta(x - 1) + c_2\delta(x + 1)$$

(e) Can use $U = \delta(x - a)$ again to make $\sin(x) = 0$

$$U = c_k\delta(x - k\pi) = \sum_{k=-\infty}^{\infty} c_k\delta(x - k\pi)$$

(f) Triple root at $x = 0$, use the relation $x^k U = 0 \rightarrow U = \sum_{k=0}^{n-1} c_k \delta^{(k)}(x)$

$$U = c_0\delta(x) + c_1\delta'(x) + c_2\delta''(x)$$

□

Problem D.9

Determine the Fourier transform of

- (a) $\delta(t - 1)$
 (b) $\delta'(t + 1)$

(c) 2

(d) $\sin(2t)$

(e) $(t+1)^1$

(f) $e^t \theta(t)$

Solution:

(a) The Fourier transform of $\delta(x)$ is $\mathcal{F}[\delta(x)] = 1$ and the translation rule is $\mathcal{F}[f(t-t_0)] = e^{-i\omega t_0} \mathcal{F}[f(t)]$

$$\mathcal{F}[\delta(t-1)] = e^{-i\omega}$$

(b) The derivative rule is $\mathcal{F}[f'(t)] = i\omega \mathcal{F}[f(t)]$. Using this and the translation rule

$$\mathcal{F}[\delta'(t+2)] = i\omega e^{2i\omega}$$

(c) $\square$